第七章 原子核物理概论.

结合能: $B(Z,A) = [Zm_p + (A-Z)m_n - m(Z,A)]c^2$

原子核质量、 $m(Z,A)$

原子质量: $M(Z,A) = m(Z,A) - Zm_e - \frac{B_e}{c^2}$

例: ^4He 的结合能

$B(4,2) = 2m_p + 2m_n - m_{He}$

比结合能: $\varepsilon(Z,A) = \frac{B(Z,A)}{A}$

Fe 左侧, 轻核聚变.

Fe 右侧: 重核裂变.

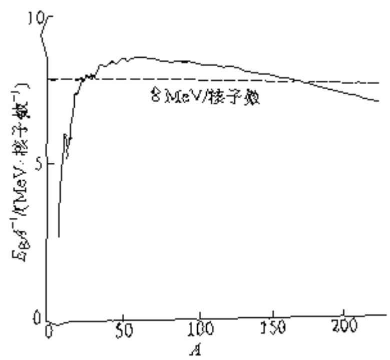


图 33.1 原子核平均结合能曲线

结合能的经验公式   液滴模型.

实验测得, $R = r_0 A^{1/3}$

$\rho = \frac{m}{V} = \frac{A/N_A}{\frac{4}{3}\pi R^3} = \frac{3}{4\pi r_0^3 N_A} = 2.3\times10^{17}\,kg\cdot m^{-3}$

原子核 ρ 为 const., ε 为 const.

$E_B = E_{BV} - E_{BS} - E_{BC}$

1. E_{BV}: 体积能

$E_{BV} = a_V A$

2. E_{BS}: 表面能

$E_{BS} = a_S A^{2/3}$

3. E_{BC}: 库仑能

$E_{BC} = \frac{3}{5}\frac{1}{4\pi\varepsilon_0}\frac{Z(Z-1)e^2}{R} = a_C Z^2 A^{-1/3}$ $(R \sim A^{1/3})$

§ 35 核矩

电子：$\mu_e = -\frac{e\hbar}{2m_e}(g_l \vec{l} + g_s \vec{s})$

$= (\vec{l} + 2\vec{s})\mu_B$

质子、中子：$\mu_p = \frac{e\hbar}{2m_p}(\vec{l} + g_{p,s}\vec{s}) = (\vec{l} + g_{p,s}\vec{s})\mu_N$

$\mu_n = \frac{e\hbar}{2m_p} g_{n,s}\vec{s} = g_{n,s}\vec{s}\,\mu_N$

核磁矩：

$$\begin{cases} \vec{\mu}_I = g_I \mu_N \vec{I} \\ \mu_{I,z} = g_I \mu_N m_I \end{cases}$$

$U = -\vec{\mu}\cdot\vec{B} = -g_I \mu_N B\, m_I$

核磁共振：

$\Delta E = g_I \mu_N B = h\nu$

例：$\nu_1 = 42.57\ \mathrm{MHz}$ 时，观测到氢的共振吸收，$g_H = 5.585$.

$\nu_2 = 16.55\ \mathrm{MHz}$ 时，观测到锂的共振吸收，锂：$I = \frac{3}{2}$ 求 g_{Li} 和 μ_{Li}

$B = \frac{h\nu_1}{g_H \mu_N} = 1T$

$g_{Li} = \frac{h\nu_2}{\mu_N B} = g_H \frac{\nu_2}{\nu_1} = 2.171$

$\mu_{Li} = g_{Li}\mu_N I = 3.257\,\mu_N$

§37 放射性衰变

1. (1) α衰变：放出两个 ${}^4_2He$

   (2) β^-衰变：放出 e^-、μ^-

   β^+衰变：放出 e^+、μ

   电子俘获：原子核俘获另外一个电子

   (3) γ衰变：放出电磁波

2. 指数衰变律

$$-dN = \lambda N dt$$

$$\Rightarrow N = N_0 e^{-\lambda t}$$

半衰期：$\frac{N}{2} = N_0 e^{-\lambda T_{1/2}} \Rightarrow T_{1/2} = \frac{1}{\lambda}\ln 2$

平均寿命：
$$\bar{t} = \frac{1}{N_0}\int_0^\infty t\, dN$$
$$= -\lambda \int_0^\infty t e^{-\lambda t} dt$$
$$= (te^{-\lambda t}\Big|_0^\infty - \int_0^\infty e^{-\lambda t} dt)$$
$$= \frac{1}{\lambda} = \frac{1}{\ln 2} T_{1/2}$$

放射性活度：
$$A = -\frac{dN}{dt} = \lambda N_0 e^{-\lambda t} = A_0 e^{-\lambda t}$$

3. 级联衰变

$$A \to B \to C$$

$$\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N_B$$
$$= \lambda_A N_{A0} e^{-\lambda_A t} - \lambda_B N_B$$

$$N_B = B e^{-\lambda_B t} \qquad \frac{dN_B}{dt} = \frac{dB}{dt} e^{-\lambda_B t} - \lambda_B B e^{-\lambda_B t}$$

$$\Rightarrow \frac{dB}{dt} e^{-\lambda_B t} - \lambda_B B e^{-\lambda_B t} = \lambda_A N_{A0} e^{-\lambda_A t} - \lambda_B B e^{-\lambda_B t}$$

$$\frac{dB}{dt} = \lambda_A N_{A0} e^{(\lambda_B - \lambda_A)t}$$

$$\Rightarrow B = \frac{\lambda_A N_A}{\lambda_B - \lambda_B} e^{(\lambda_B - \lambda_A)t} + C, \quad N_B|_{t=0} = 0 \Rightarrow C = \frac{\lambda_A N_A}{\lambda_A - \lambda_B}$$

可得 $N_B = \frac{\quad}{\lambda_A - \lambda_B} e \quad - e$

* 当子核寿命 << 母核寿命，$\lambda_A << \lambda_B$

$t >> \frac{1}{\lambda_B}$ 时，$N_B = \frac{\lambda_A N_{A0}}{\lambda_B - \lambda_A} e^{-\lambda_A t}$ 久期平衡：按母核衰变规律衰变.

$\lambda_A N_A \approx \lambda_B N_B$

4. 同位素生产.

$$\frac{dN}{dt} = P - \lambda N$$

$N = A e^{-\lambda t}$，$\frac{dN}{dt} = \frac{dA}{dt} e^{-\lambda t} - \lambda A e^{-\lambda t}$

$\frac{dA}{dt} = P e^{\lambda t} \rightarrow A = \frac{P}{\lambda} e^{\lambda t} + C$

$N|_{t=0} = (\frac{P}{\lambda} + C) = 0 \rightarrow C = -\frac{P}{\lambda}$

解得 $N = \frac{P}{\lambda}(1 - e^{-\lambda t})$

§ 38 α衰变.

$${}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + \alpha$$

能量守恒：$m_X c^2 = m_Y c^2 + m_\alpha c^2 + E_\alpha + E_r$

衰变能 $Q_\alpha = E_\alpha + E_r = (m_X - m_Y - m_\alpha)c^2$

假设母核静止：

$$m_Y v_Y = m_\alpha v_\alpha$$

$$Q_\alpha = E_\alpha + E_r = \frac{1}{2} m_\alpha v_\alpha^2 + \frac{1}{2} m_Y v_Y^2$$

$$= E_\alpha \left(1 + \frac{m_\alpha}{m_Y}\right) = \frac{A}{A-4} E_\alpha$$

§39 β衰变

1. 中微子

$Q_\beta = E_e + E_\nu + E_r$

β^-：$n \to p + e + \bar{\nu}_e$

$^3H \to {}^3He + e + \bar{\nu}_e$

β^+：$p \to n + e^+ + \nu_e$

$^{13}N \to {}^{13}C + e^+ + \nu_e$

电子俘获（EC）：

$^7Be + e_K^- \to {}^7Li + \nu_e$

2. β^-衰变

$^A_ZX \to {}^A_{Z+1}Y + e^- + \bar{\nu}_e$

衰变能：$Q_{\beta^-} = (m_X - m_Y - m_e)c^2 = (M_X - M_Y)c^2$

3. β^+衰变

$^A_ZX \to {}^A_{Z-1}Y + e^+ + \nu_e$

衰变能：$Q_{\beta^+} = (m_X - m_Y - m_e)c^2$

$= (M_X - M_Y - 2m_e)c^2$

4. 电子俘获（EC）

$^A_ZX + e_i^- \to {}^A_{Z-1}Y + \nu_e$

衰变能：$Q_{EC} = (m_X + m_e - m_Y)c^2 - W_i$

$= (M_X - M_Y)c^2 - W_i$

§ 40 γ衰变

穆斯堡尔效应：

$$\begin{cases} E_0 = h\nu + E_R \\ \dfrac{h\nu}{c} = p_\gamma = p_R \end{cases}$$

放出光子：$E_{\gamma e} = E_0 - \dfrac{(h\nu)^2}{2mc^2}$

吸收光子：$E_{\gamma a} = E_0 + \dfrac{(h\nu)^2}{2mc^2}$

能级宽度：$\Gamma = \dfrac{\hbar}{\tau}$，$\tau$为寿命.

$2E_R < \Gamma \Rightarrow$ 可以发生共振吸收.

§ 41 核反应

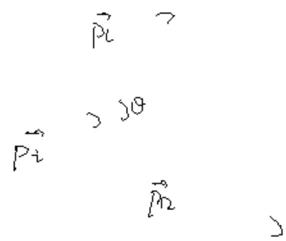
1. Q方程

$i + T \rightarrow l + R$

$$m_i c^2 + K_i + m_T c^2 + K_T = m_l c^2 + K_l + m_R c^2 + K_R$$

$$Q = (m_i + m_T - m_l - m_R)c^2$$

$$= K_l + K_R - K_T - K_i = K_l + K_R - K_i$$



动量守恒：

$\vec{p_i} = \vec{p_l} + \vec{p_R}$ $\quad \vec{p_T} = 0$，靶核静止.

$$p_R^2 = p_i^2 + p_l^2 - 2p_i p_l \cos\theta$$

$$2m_R K_R = 2m_i K_i + 2m_l K_l - 2\sqrt{m_i m_l K_i K_l}\cos\theta$$

$$\Rightarrow \quad Q = \left(1+\frac{m_l}{m_R}\right)K_l - \left(1-\frac{m_i}{m_R}\right)K_i - \frac{2\sqrt{m_i m_l K_i K_l}}{m_R}\cos\theta$$

在质心系中

$$K_i' + K_T' = \frac{1}{2} m_i (v_i - v_c)^2 + \frac{1}{2} m_T v_c^2$$

$$= \frac{1}{2} m_i v_i^2 \left(\frac{m_T}{m_i+m_T}\right)^2 + \frac{1}{2} m_T \left(\frac{m_i}{m_i+m_T}\right)^2 v_i^2$$

$$= \frac{m_i m_T v_i^2}{2(m_i+m_T)} = \frac{m_T}{m_i+m_T} K_i$$

反应后，产物在质心系中可以静止.

以反应能Q为质心系的最小动能

$$K_i \geq \frac{m_i+m_T}{m_T}|Q|$$

例：1. 用多大能量的质子轰击固定氚靶才能发生反应

$$p + t \to n + {}^3He$$

2. 若入射质子的能量为 3 MeV，发射中子和入射方向成 90° 角，求出射中子和 ^3He 的动能

1、 $Q = m_p + m({}^3H) - m_n - m({}^3He) = -0.762\ MeV$

入射最小动能：$T_{p\,min} = \dfrac{m_p + m_{^3H}}{m_{^3H}} |Q| =$

2、 $Q = K_n + K_{^3He} - K_p$

动量守恒：$\vec{p}_p = \vec{p}_n + \vec{p}_{^3He} \to p_{^3He}^2 = p_p^2 + p_n^2$

$$2m_{^3He} K_{^3He} = 2m_p K_p + 2m_n K_n$$

解得 $K_{^3He} = \dfrac{m_p}{m_{^3He}} K_p + \dfrac{m_n}{m_{^3He}} K_n$

$$Q = K_n\left(1 + \frac{m_n}{m_{^3He}}\right) - K_p\left(1 - \frac{m_p}{m_{^3He}}\right)$$

得 $K_n =$

$K_{^3He} =$